From Monolith to Microservices: A Classification of Refactoring Approaches

Source: J. Fritzsch *et al.* "From Monolith to Microservices: A Classification of Refactoring Approaches," *Proceedings of the First International Workshop on Software Engineering Aspects of Continuous Development and New Paradigms of Software Production and Deployment (DEVOPS 2018)*, 2019. [Online]. Available: http://link.springer.com/10.1007/978-3-030-06019-0_10

Paper Information: Literature Review (LR)

Rating: ⭐ ⭐ ⭐

Reading Progress: Abstract, Was only mentioned by others

Introduction (What is the Paper about?)

- **Identify and classify existing refactoring approaches in the context of monolith decomposition**
  - outlines existing approaches using a decision guide

Methods

- **Literature Review** (no SLR)
  - compared 10 existing studies
  - evaluated selected methods along *granularity*, *input and output types*, *result evaluation*, *tool support* and *validation*

Scientific Benefit

- **Classification**
  - *Static Code Analysis (SCA)*
  - *Meta-data aided (MDA)*
  - *Workload-data aided (WDA)*
  - *Dynamic Microservice Composition (DMC)*

Challenges / Future Work

Limitations

Paper Outline and Read Progress

Own Comments

- (-) does not cover microservice decomposition methods (abgaz2023Decomposition)
- (-) broad scope since it includes *greenfield microservices development* along with *monolith decomposition* (abgaz2023Decomposition)